Quantifying Epistemic Uncertainty in Diffusion Models

Fisher–Laplace Randomized Estimation (FLARE), AISTATS'26

Aditi Gupta[1] Raphael A. Meyer[2,3] Yotam Yaniv[1] Elynn Chen[4] N. Benjamin Erichson[1,2]

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[1]Lawrence Berkeley National Lab, Berkeley, USA
[2]International Computer Science Institute, Berkeley, USA
[3]University of California, Berkeley, USA
[4]New York University, New York, USA

Generative AI

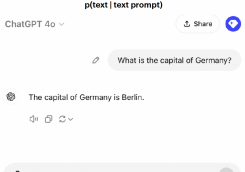
- Generative models now power high-impact applications: vision, language, time-series forecasting, scientific simulation surrogates, and drug discovery.

p(image | text prompt)



Instruction: Teddy bear on a skateboard on the moon. (Generated with DALL·E 3.)

p(text | text prompt)



p(code | text prompt)

```python
import numpy as np

def wave_2d_solver(Lx, Ly, Nx, Ny, c, dt, T):
    dx, dy = Lx / (Nx - 1), Ly / (Ny - 1)  # Grid spacing
    Nt = int(T / dt)  # Time steps
    C_x, C_y = c * dt / dx, c * dt / dy

    if C_x >= 1 or C_y >= 1:
        raise ValueError("Unstable scheme: decrease dt or increase dx/dy.")

    # Initialize grids
    u = np.zeros((Nx, Ny))
    u_prev, u_next = np.zeros((Nx, Ny)), np.zeros((Nx, Ny))

    # Initial condition: Gaussian pulse in the center
    x, y = np.linspace(0, Lx, Nx), np.linspace(0, Ly, Ny)
    X, Y = np.meshgrid(x, y)
    u = np.exp(-((X - Lx/2)**2 + (Y - Ly/2)**2) / 0.1)
    u_prev = np.copy(u)

    # Time-stepping loop
    for _ in range(Nt):
        u_next[1:-1, 1:-1] = (2 * u[1:-1, 1:-1] - u_prev[1:-1, 1:-1] +
                              C_x**2 * (u[2:, 1:-1] - 2 * u[1:-1, 1:-1] + u[:-2, 1:-1]) +
                              C_y**2 * (u[1:-1, 2:] - 2 * u[1:-1, 1:-1] + u[1:-1, :-2]))
        u_prev, u = u, u_next

    return u
```

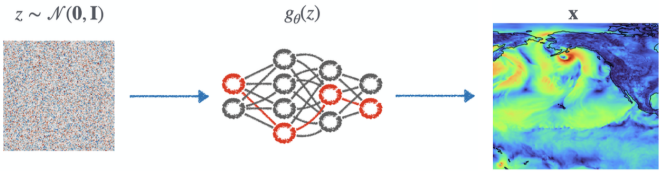
Instruction: Implement a solver for the 2-D wave equation. (Generated with GPT-4.)

- Despite these advances, current generative models, such as diffusion models, lack principled mechanisms to quantify uncertainty in generated samples, limiting their reliability for scientific and decision-making tasks.

What is uncertainty in generative models?

- Generative models map noise to data:

$$x_0 = g_\theta(z), \quad z \sim \mathcal{N}(0, I)$$



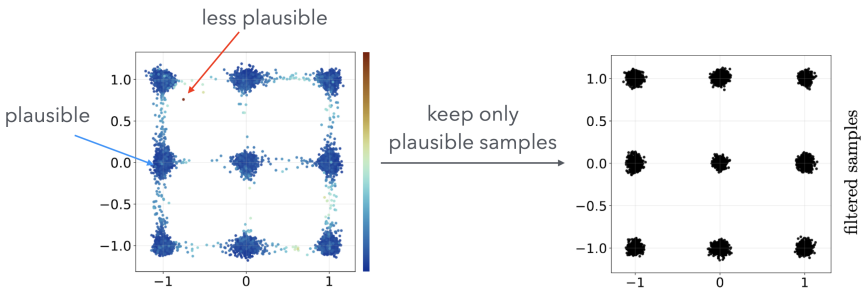
- Outputs vary due to randomness in z and stochasticity during inference

But when is the model actually uncertain?

Small changes in input $\rightarrow$ very different outputs e.g., regions with little data or off-manifold inputs

Motivating Example

- Considering a toy dataset of 2D Gaussian points with 9 modes
- We want a metric that differentiates off-manifold points



Predictive variance alone is not enough — we need uncertainty that is interpretable and enables filtering

Why do models differ in output?

Total Predictive Uncertainty

$$\underbrace{\mathrm{Var}[x]}_{\text{total}} = \underbrace{\mathbb{E}_\theta[\mathrm{Var}(x|\theta)]}_{\text{aleatoric}} + \underbrace{\mathrm{Var}_\theta[\mathbb{E}(x|\theta)]}_{\text{epistemic}}$$

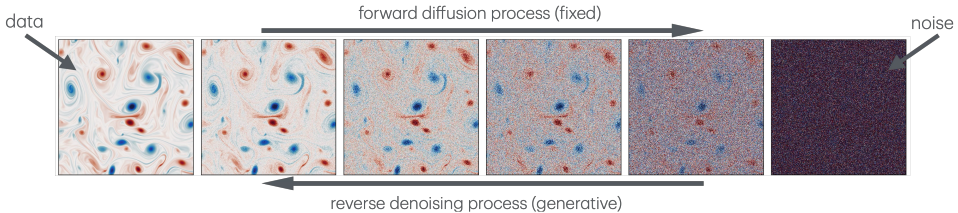
Captures both data randomness and model uncertainty.

Intuition

Quantify where different models disagree on what the output should be

Epistemic uncertainty or uncertainty in θ tells us where the model is unreliable

Diffusion and Uncertainty



- Start from noise and iteratively denoise to generate samples
- Each reverse step depends on a learned model

$$x_{t-1} = a_t x_t - b_t \epsilon_\theta(x_t, t) + \eta_t$$

Key Question

How does uncertainty in θ affect the generated sample?

From Model Parameters to Outputs

Reverse step:

$$x_{t-1} = a_t x_t - b_t \varepsilon_\theta(x_t, t) + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \tilde{\beta}_t I).$$

Proposition (One-step Fisher–Laplace separation)

Fix a diffusion noise path η and assume a local Gaussian posterior $\theta \sim \mathcal{N}(\hat{\theta}, \Sigma_\theta)$. Then (treating x_t as fixed at its realized value along the path):

$$\underbrace{\mathrm{Cov}_\theta\big(x_{t-1}(\theta,\eta) \,\big|\, x_t, t, \eta\big)}_{\text{epistemic}} \;\approx\; b_t^2 J_t \Sigma_\theta J_t^\top, \qquad J_t = \nabla_\theta \varepsilon_\theta(x_t, t)\big|_{\hat{\theta}}.$$

epistemic covariance

Jacobian

posterior covariance

$\Sigma_\theta \approx (H_\theta + \lambda I)^{-1}$

local Laplace approximation

$H_\theta \approx J^T J$

empirical Fisher

- Condition on diffusion randomness to isolate parameter-driven uncertainty

Epistemic Uncertainty Propagation

- Epistemic uncertainty propagates through the reverse step via the model

Usable Recursion:

$$\Sigma_{t-1}^{\mathrm{ep}} \;\approx\; a_t^2\,\Sigma_t^{\mathrm{ep}} + b_t^2\,J_t \Sigma_\theta J_t^\top$$

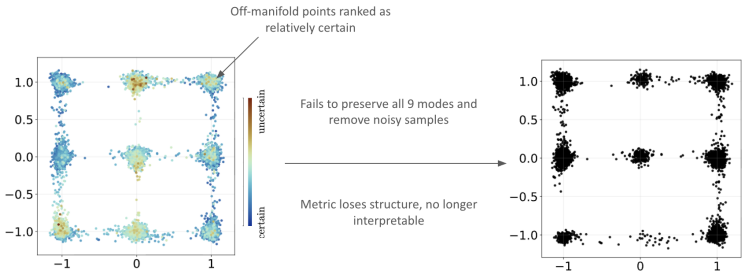
- First term: propagated past uncertainty
- Second term: new uncertainty injected through the model

Unrolling across all steps gives the total epistemic covariance:

$$\Sigma_0^{\mathrm{ep}} = \sum_{s=1}^{T} \left(\prod_{j<s} a_j^2 \right) b_s^2\, J_s\, \Sigma_\theta\, J_s^\top .$$

Why is this challlenging to compute?

- Σ_θ requires computing the Hessian $\Rightarrow$ **intractable for modern networks**
- Common workaround: **last-layer Laplace (LLLA)** used in existing methods such as BayesDiff



LLLA is efficient but no longer interpretable

$\rightarrow$ We need a scalable, network-wide approximation

Least-Squares View of FLARE

Idea. Rewrite our object $V = J_t(J_{\text{pop}}^\top J_{\text{pop}})^+ J_t^\top$ as solution to a least-squares optimization problem

Lemma (Least-Squares Equivalence)

Let $A \in \mathbb{R}^{p\times k}$, $B \in \mathbb{R}^{p\times \ell}$, and $X_\star = \arg\min_X \|AX - B\|_F^2$. Then:

$$B^\top (AA^\top)^+ B = X_\star^\top X_\star.$$

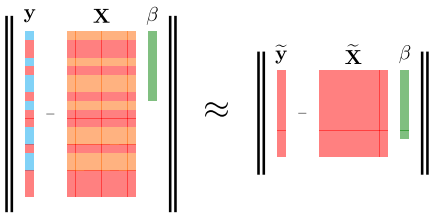
Interpretation.

- $V = X_\star^\top X_\star$ with $A = J_{\text{pop}}^\top$, $B = J_t^\top$.

- Enables us to use a principled randomized subnetwork approximation

Random Subnetwork Sampling as Sketching

Idea. Sampling a random subset of network parameters is equivalent to applying a **sketching operator** that selects columns of the Jacobian.



- The idea is to select and reweight a subset of rows to form a smaller problem whose solution approximates the full least-squares estimate.
- Guarantees need mild assumptions, which are not violated in well-trained deep network settings.

Theoretical Guarantee for FLARE

Goal: Quantify how close the random subnetwork Laplace is to the full Fisher–Laplace covariance.

Theorem (Trace Error Bound)

For well-conditioned, low-coherence Jacobians J_{pop} and J_t, a random subnetwork of size $m \ll p$ satisfies

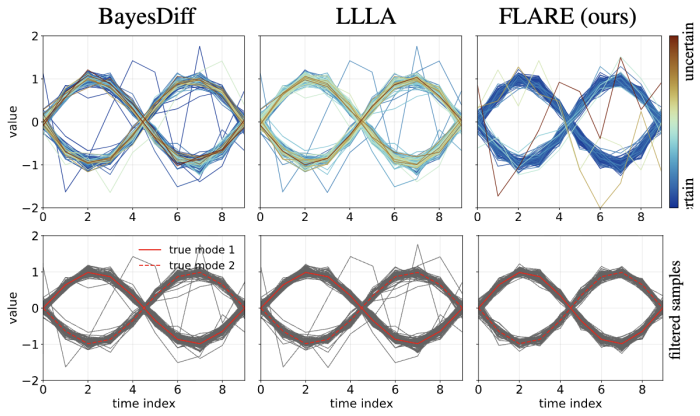
$$\left|\operatorname{tr}(V - \tilde{V})\right| \leq \tilde{\mathcal{O}}\left(\sqrt{\frac{p\mu}{m\delta}}\,\frac{\kappa}{\gamma}\right)\operatorname{tr}(V),$$

with probability $\geq 1 - \delta$.

Implications:

- Error $\propto 1/\sqrt{m}$ — larger subnetworks $\Rightarrow$ tighter approximation.
- Holds under mild conditioning, coherence, and alignment assumptions.

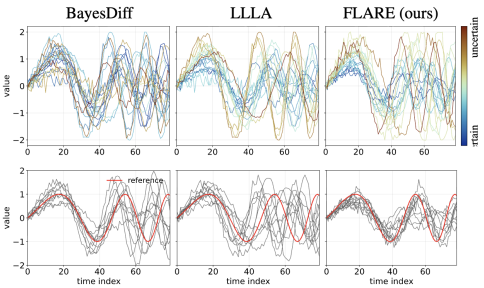
Qualitative Results: Time-Series



(a) Sinusoidal time series dataset.

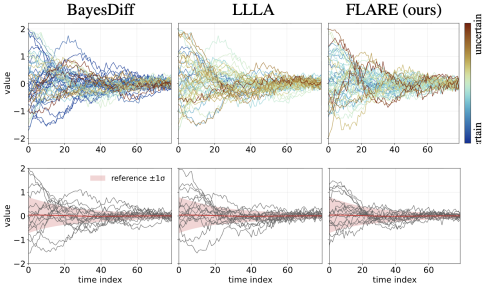
- Rank samples by $\mathrm{tr}(\Sigma_0^{\mathrm{ep}})$; keep lowest-$p$%

Time Series Results (cont'd)



(b) Chirp time series dataset.

Unreasonable samples are highlighted



(c) Damped Sinusoidal time series dataset.

Chirp signal: sharper boundaries, uncertain transitions captured

Observation

Epistemic heatmaps align with regions of low density or mode transitions, capturing model sensitivity rather than data noise.

Quantitative Results

Comparison of uncertainty-based filtering methods across two datasets. Higher gap-closure and lower ROC-AUC indicate improved alignment between filtered generations and the training data. Best values are highlighted in bold.

Dataset	Method	Gap-Closure (%)	ROC-AUC ↓	p (bootstrap)
Sines	BayesDiff	+13.3654	0.6153	0.0001
	LLLA	+47.1873	0.5814	0.0201
	FLARE (ours)	**+93.0814**	**0.5003**	**0.0012**
Chirp	BayesDiff	+41.7337	0.6616	0.0030
	LLLA	+59.0583	0.5891	0.0116
	FLARE (ours)	**+74.3137**	**0.5345**	**0.0002**
Damped Sines	BayesDiff	+10.3984	0.6861	0.0001
	LLLA	-18.7540	0.7754	0.0066
	FLARE (ours)	**+85.0**	**0.5085**	**0.0002**

Takeaways and Future Work

Takeaways

- Epistemic uncertainty captures **model uncertainty** and enables **interpretable filtering**
- Existing approximations (e.g., LLLA) fail to capture meaningful structure
- **FLARE** provides a **scalable, network-wide approximation** with strong empirical performance

Future Directions

- Scaling to larger and conditional generative models
- Understanding approximation quality and stability
- Leveraging uncertainty for filtering, calibration, and robustness

Assumptions & Practical Notes

- **Well-conditioned curvature:** Moderate condition number $\kappa(J_{\text{pop}})$ and low coherence μ ensure stable randomized sketches. (Wider nets or dropout help.)
- **Alignment assumption:** $\|J_t P\|_F \geq \gamma \, \|J_t(I - P)\|_F$, where P projects onto the row space of J_{pop}. Ensures that sampled subnets capture relevant directions.
- **Regularization:** Small ridge λ used for numerical stability; theory considers $\lambda \to 0$.
- **Sampling:** Uniform column sampling suffices; leverage-score sampling is a drop-in upgrade.
- **Computation:** Cost scales linearly with sampled rank m, independent of full parameter count p.

Backup

Parameter Budget Ablation

- Retaining different fractions of parameters: $\{1\%, 5\%, 10\%, 30\%, 50\%\}$
- Stable behavior already at 10–30%, diminishing returns beyond

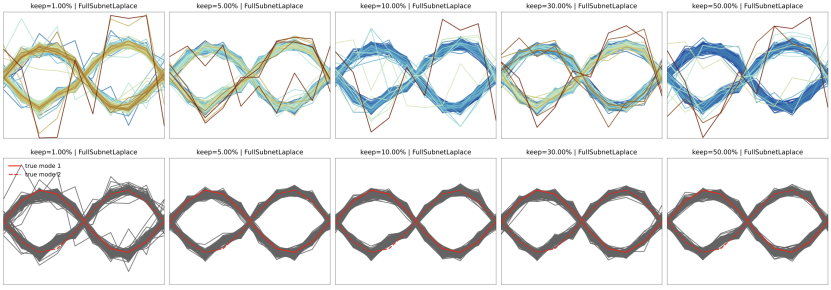


Figure 5: Parameter–budget ablation with FullSubnetLaplace. We retain $\{1\%, 5\%, 10\%, 30\%, 50\%\}$ of parameters. As the kept fraction increases, trajectories tighten around the data manifolds and the epistemic covariance $b_t^2 \mathbf{J}_t \boldsymbol{\Sigma}_\theta \mathbf{J}_t^\top$ contracts smoothly; 10–30% already yields stable mode coverage, with diminishing returns beyond 30–50%.

Full-Curvature Regime: Four-Way Comparison

- In small networks, full Hessian is tractable — allows direct comparison to ground truth
- Four methods compared: BayesDiff, Full-Hessian Laplace, BayesDiff (epistemic-only), FLARE

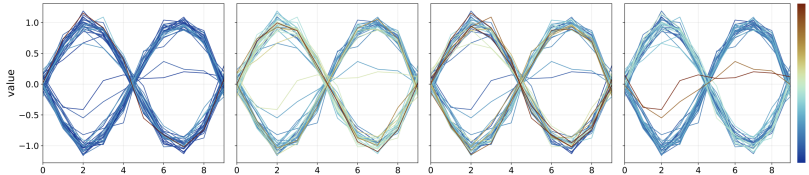


Figure 6: **Four-way epistemic UQ comparison in the full-curvature regime.** We compare uncertainty estimates from (left to right): *BayesDiff (last-layer Laplace, total recursion)*, *Fuller-Hessian Laplace (Bayesdiff recursion)*, *BayesDiff (epistemic-only recursion)*, and *FLARE (Fisher–Laplace projection)*. In this controlled setting, full-curvature computation is feasible and allows direct inspection of epistemic structure. The scalable approximation of Kou et al. (2024) (BayesDiff) already breaks down in this regime, producing uncertainty maps that are not epistemically meaningful relative to the full-Hessian reference, whereas FLARE closely tracks the curvature-based baseline while avoiding full-Hessian computation.

Propagation of Epistemic Uncertainty: Full Recursion

Epistemic covariance recursion (exact form)

$$\Sigma_{t-1}^{\mathrm{ep}} = a_t^2\,\Sigma_t^{\mathrm{ep}} + b_t^2\,J_t\Sigma_\theta J_t^\top \;-\; 2a_t b_t\,\underbrace{\mathrm{Cov}_\theta\big(x_t(\theta),\,J_t(\theta-\hat{\theta})\big)}_{\text{cross-covariance term}}.$$

- First term: propagation of past uncertainty
- Second term: new uncertainty from the model

Dominant contribution: $b_t^2\,J_t\Sigma_\theta J_t^\top$

Why the Cross Term Can Be Controlled

Bound (absorbability).

Using Cauchy–Schwarz and Young's inequality, for any $\delta > 0$:

$$2|a_t b_t|\,|\mathrm{Cov}(X, Y)| \;\le\; \delta\, a_t^2\, \mathrm{Var}(X) + \delta^{-1}\, b_t^2\, \mathrm{Var}(Y),$$

with

$$X = x_t(\theta), \qquad Y = J_t(\theta - \mathbb{E}\theta).$$

The cross term cannot dominate the recursion: it is bounded by the same two variance terms already present.

Why we drop it.

- **Assumption (local decoupling):** Near $\hat{\theta}$, fluctuations in $x_t(\theta)$ and in $J_t(\theta - \hat{\theta})$ are weakly correlated.
- **Empirical check:** Monte Carlo estimates show the cross term is orders of magnitude smaller than the main Fisher term in our experiments. (typically 2 orders of magnitude)

Resulting approximation used in FLARE:

$$\Sigma_{t-1}^{\mathrm{ep}} \;\approx\; a_t^2\, \Sigma_t^{\mathrm{ep}} + b_t^2\, J_t \Sigma_\theta J_t^\top.$$

Cross-Covariance Term: Empirical Validation

- Ran full epistemic recursion with and without Monte Carlo estimate of the cross term
- Relative change $\Delta u / u^{\text{noX}}$ reported across $N = 80$ trajectories

Dataset	Mean (%)	Std (%)	Max (%)	SE (%)
Sine	0.00338	0.00440	0.02220	0.00049
Chirp	0.00495	0.00610	0.02840	0.00068

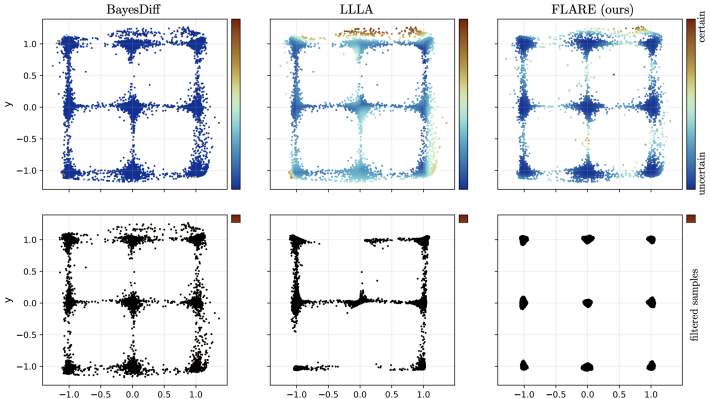
Takeaway

Cross-covariance term contributes less than $10^{-2}\%$ to the epistemic score. Dropping it has no practical effect — and saves substantial compute.

DDIM Results

- FLARE extends naturally to DDIM (deterministic sampler, $\eta_t \equiv 0$)
- Aleatoric term $\tilde{\beta}_t I$ vanishes; epistemic term remains

$$\mathrm{Cov}(x_{t-1}|x_t, t) \approx b_t^2 J_t \Sigma_\theta J_t^\top$$



BayesDiff vs. FLARE: Key Differences

	BayesDiff	**FLARE (ours)**
What it measures	Total predictive variance — mixes aleatoric + epistemic	Pure epistemic variance — parameter uncertainty only
Noise path	Averaged over diffusion randomness	Conditioned on fixed noise path η
Laplace scope	Last layer only (LLLA)	Random subnetwork across all layers
Cross terms	Propagates predictive cross term $\mathrm{Cov}(x_t, \varepsilon_t)$	Drops cross term (shown negligible)
Guarantee	None for approximation quality	$O(1/\sqrt{m})$ trace error bound

Fundamentally different objects — not just an engineering improvement